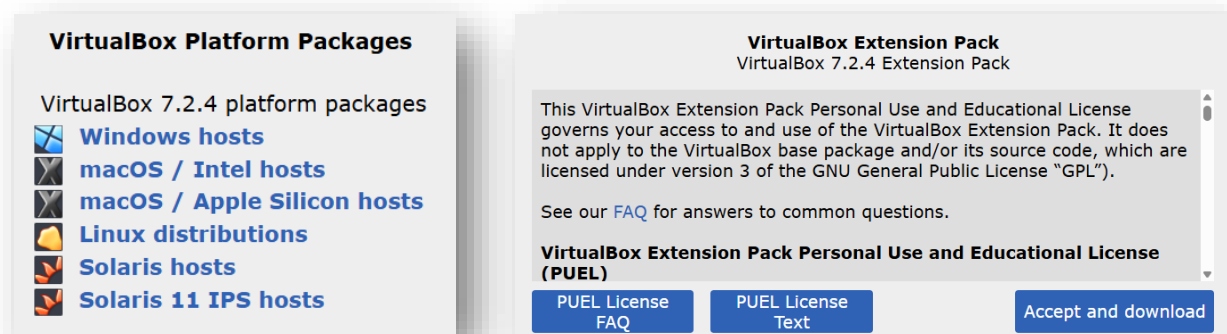


Lab0: VirtualBox and Linux Installation – The Linux Terminal

1. VirtualBox

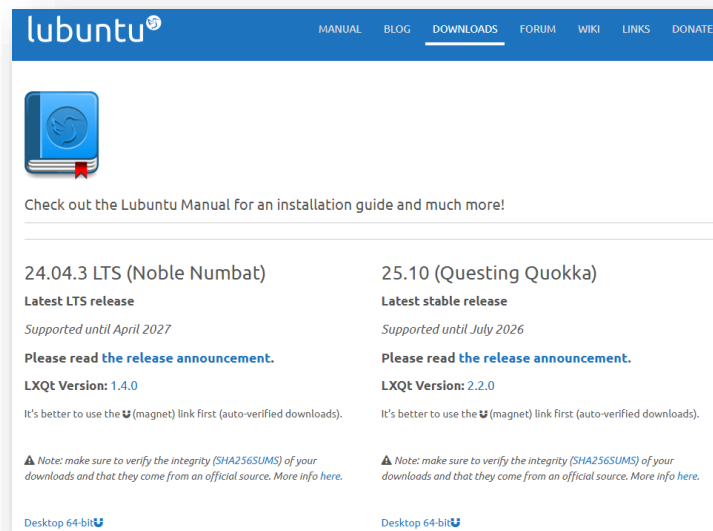
Go to the official VirtualBox website (<https://www.virtualbox.org/wiki/Downloads>) and download “VirtualBox” as well as “VirtualBox Extension Pack”.



Install VirtualBox and the VirtualBox Extension Pack (in this order).

2. Lubuntu

Go to the official Ubuntu website (<https://lubuntu.me/downloads/>) and download the



3. The Linux Terminal

Exercise 1: Basic Navigation and File Operations

Please use basic commands like ls, cd, pwd, mkdir, touch, cp, mv, and rm to achieve several goals:

a. List Files and Directories:

Open the terminal and navigate to your home directory. What is the name of this directory? What are the two possible commands to go to your home directory directly from any location in your filesystem?

Use ls to list all files and directories.

Use ls -l to list files in long format. What information are present on each line? What are the rights/privileges on the files/directory?

b. Create and Navigate Directories:

Use mkdir to create a directory named "practice".

Use cd to enter the "practice" directory.

Use pwd to print the current working directory.

c. Create and Manipulate Files:

Use touch to create a file named file1.txt.

Use cp to copy file1.txt to file2.txt.

Use mv to rename file2.txt to renamed_file2.txt.

Use rm to delete renamed_file2.txt.

Exercise 2: Viewing and Editing Files

Use these commands cat, less, echo, and nano to answer the following questions:

a. View File Contents:

Use echo "Hello, Linux!" > file1.txt to add text to file1.txt.

Use cat file1.txt to display the contents of file1.txt.

Use less file1.txt to view the file page by page.

b. Edit Files:

Use vi file1.txt to open the file in the vi editor. Add a new line of text, save the file, and exit.

Use nano file2.txt to open the file in the nano editor. Add a new line of text, save the file, and exit.

Exercise 3: Using Pipes, Learn to use pipes (|) to chain commands.

a. List and Filter Files

Use ls -l | less to view the long listing of files page by page.

Use ls -l | grep ".txt" to filter and display only .txt files.

b. Count Words and Lines:

Use cat file1.txt | wc -l to count the number of lines in file1.txt.

Use cat file1.txt | wc -w to count the number of words in file1.txt.

Exercise 4: Redirection, Learn to redirect output using >, >>, and <.

a. Redirect Output:

Use echo "New line of text" > newfile.txt to create newfile.txt with the text.

Use echo "Appended text" >> newfile.txt to append text to newfile.txt.

b. Redirect Input:

Use grep "Linux" < newfile.txt to search for the word "Linux" in newfile.txt.

Exercise 5: Understanding Paths, Learn to use absolute and relative paths.

a. Absolute Paths:

Use cd /home/username/practice to navigate to the practice directory using an absolute path.

Use ls /home/username/practice to list files in the practice directory.

b. Relative Paths:

Navigate to your home directory and use cd practice to enter the practice directory using a relative path.

Use ls ../ to list files in the parent directory.

How can you know if a path is absolute or relative?

Exercise 6: Combining Paths and Commands

Objective: Use paths with commands like cp, mv, and rm.

a. Copy Files Using Paths:

Use `cp /home/username/practice/file1.txt /home/username/` to copy file1.txt to your home directory.

b. Move Files Using Paths:

Use `mv /home/username/file1.txt /home/username/practice/renamed_file.txt` to move and rename the file.

c. Remove Files Using Paths:

Use `rm /home/username/practice/renamed_file.txt` to delete the file.

Bonus Challenge: Combining Everything, Learn how to combine navigation, pipes, and paths.

a. Create a Directory Structure:

Create directories dir1, dir2, and dir3 inside practice.

Create a file notes.txt in each directory with some text containing several times the word "Linux".

b. Search and Filter:

Use `grep "Linux" /home/username/practice/*/notes.txt` to search for the word "Linux" in all notes.txt files.

c. Count Matches:

Use `grep -c "Linux" /home/username/practice/*/notes.txt` to count occurrences of "Linux" in each file.

Congratulations, you have learn a lot on VirtualBox, Linux Lubuntu and the use of the terminal!

Grattis, du har lärt dig mycket om VirtualBox, Linux, Lubuntu och användningen av terminalen!

Félicitations, vous avez appris beaucoup sur VirtualBox, Linux, Lubuntu et l'utilisation du terminal !